1. (10 pts) Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by
$$f(x, y, z) = x^2 + y^2 + z^2.$$
   (a) Determine the domain and range of f.
   (b) Describe geometrically the level set formed by $f(x, y, z) = 100$.

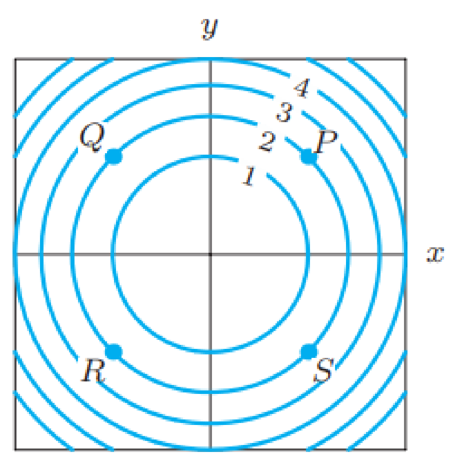
2. (10 pts)
   (a) Compute the following limit or show that the limit does not exist.
$$\lim_{(x,y)\to(0,0)} \frac{x + y^2}{2x + y}.$$
   (b) Is the following function continuous along $y = 0$?
$$f(x, y) = \begin{cases} 1 - x & y \geq 0 \\ -2 & y < 0 \end{cases}$$

3. (10 pts) Let $g(x, y, z) = x^2 - 2xy^2 + az - a$
   (a) Find an equation of the tangent plane to g at the point $(1, 1, 1)$.
   (b) For which value of a does the tangent plane pass through the origin?

4. (10 pts) Below is a contour diagram of $f(x, y)$. In each of the following cases, list the marked points in the diagram (there may be none or more than one) at which

   (a) $f_x < 0$ (b) $f_y > 0$
   (c) $f_{xx} > 0$ (d) $f_{yy} < 0$

   Hint: what kind of function $f(x, y)$ would produce a contour plot like the one shown? Compute its f_x, f_y, f_{xx} and f_{yy}.



5. (10 pts) Let $F(x) = \int_0^x \cos(\pi t)dt$.
   (a) Evaluate $F(1)$.
   (b) For what values of x is $F(x)$ positive? negative?

6. Let $T(x, y) = e^{-(x^2+y^2)}$ represent the heat of the point (x, y).

   (a) In which direction should a heat-seeking bug move from the point $(-1, 1)$ to increase its temperature fastest?

   (b) Find the directional derivative of T at the point $P = (-1, 1)$ in the direction of $(1, 1)$.

   (c) Does T have a maximum or a minimum? At where? Hint: You don't need to compute second-order derivatives. Consider that exp is monotone increasing and how that affects the extrema of T.

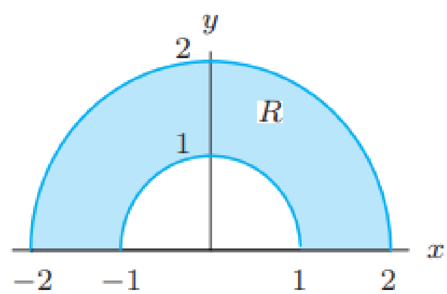
7. (10 pts) Consider the function

$$f(x, y) = |xy|.$$

   (a) Is f differentiable at $(1, 0)$? Explain.

   (b) Is f differentiable at $(0, 0)$? Explain.

8. (10 pts) Use polar coordinate to evaluate $\int_R \sqrt{x^2 + y^2}\, dA$ where R is given in the following figure.



9. (10 pts) Evaluate the following integral by using cylindrical coordinate or spherical coordinate:

$$\iiint_W \frac{z}{(x^2 + y^2)^{3/2}}\, dV, \quad W = \{(x, y, z) \mid 1 \le x^2 + y^2 \le 4,\ 0 \le z \le 4\}.$$

10. (10 pts) Decide whether the following statements are true or false.

   (a) If a function is differentiable, it must be continuous.

   (b) $\int_0^1 \int_0^x f(x, y) dy dx = \int_0^1 \int_0^y f(x, y) dx dy$.

   (c) If $\int_R f\, dA = 0$, then f is zero at all points in R.

   (d) If f and g are two functions continuous on a region R, then

$$\int_R f \cdot g\, dA = \int_R f\, dA \cdot \int_R g\, dA.$$

   (e) A local maximum of f can only occur at the critical points.